

4 Let $w_r = r(r+1)(r+2)\dots(r+9)$.

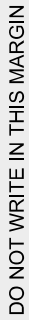
(a) Show that

$$w_{r+1} - w_r = 10(r+1)(r+2) \dots (r+9). \quad [2]$$

[illegible]

(b) Given that $u_r = (r+1)(r+2)\dots(r+9)$, find $\sum_{r=1}^n u_r$ in terms of n . [3]

[illegible]



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